

An abstract network diagram composed of dark brown circular nodes of varying sizes connected by thin, dark brown lines. The nodes are scattered across the left and top portions of the image, creating a complex web of connections. The lines vary in length and orientation, some forming triangles and others more irregular shapes. The overall effect is a sense of interconnectedness and data flow.

NOVA MART

**MARKETING ANALYTICS CAPSTONE • CAMPAIGN ROI,
CUSTOMER SEGMENTATION,
LEAD CONVERSION & CUSTOMER GROWTH**

Group 5 - Kynaa & Chaitanya

WHAT THIS MEMO COVERS

01

Data Audit Summary

What we found and how we fixed it

05

Campaign Clustering

Which campaigns to scale, maintain, or stop

02

Business KPI Dashboard

Overall health of the marketing funnel

06

Customer Segmentation

Three distinct segments and how to treat them

03

Channel & Region Performance

Where leads, conversions, and revenue come from

07

Lead Conversion Prediction

4-model comparison + priority scoring

04

Discounting Analysis

Does discount drive conversion and at what cost?

08

Repeat Buyer Prediction

Who comes back and what predicts it

DATA AUDIT: WHAT WE FOUND

5 tables · 34 documented issues · All treated before analysis

Customers	18 duplicate IDs; ages <18 or >85 imputed by region median; gender & region labels standardised
Campaigns	Mixed currency formats (\$, USD, commas); 1 negative spend → NaN; 2 impossible CTR rows flagged
Leads	Text-coded discounts ('fifteen'); '?' lead scores → NaN; mixed boolean converted 30d normalised
Sessions	Mixed True/False/Y/N booleans normalised; negative & extreme time on site (>2 hrs) → NaN
Transactions	Revenue capped at 99th pctile (\$189); 17 negative revenues → NaN; word-coded discounts fixed

⚠ Join success rates: customers↔leads 94% · customers↔transactions 59% · campaigns↔sessions 77% (orphaned records excluded from modelling)

BUSINESS KPI DASHBOARD

Baseline metrics across the full marketing funnel

5,224

Total Leads

36.8%

Conversion Rate

\$69.23

Avg CPA (Converted)

\$54.71

Avg Order Value

51.6%

Repeat Buyer Rate

8.1%

Return Rate

ROAS

2.43×total revenue / total campaign spend

Avg Days to Convert

12.4 days from lead to first purchase

Lead-to-Revenue Ratio

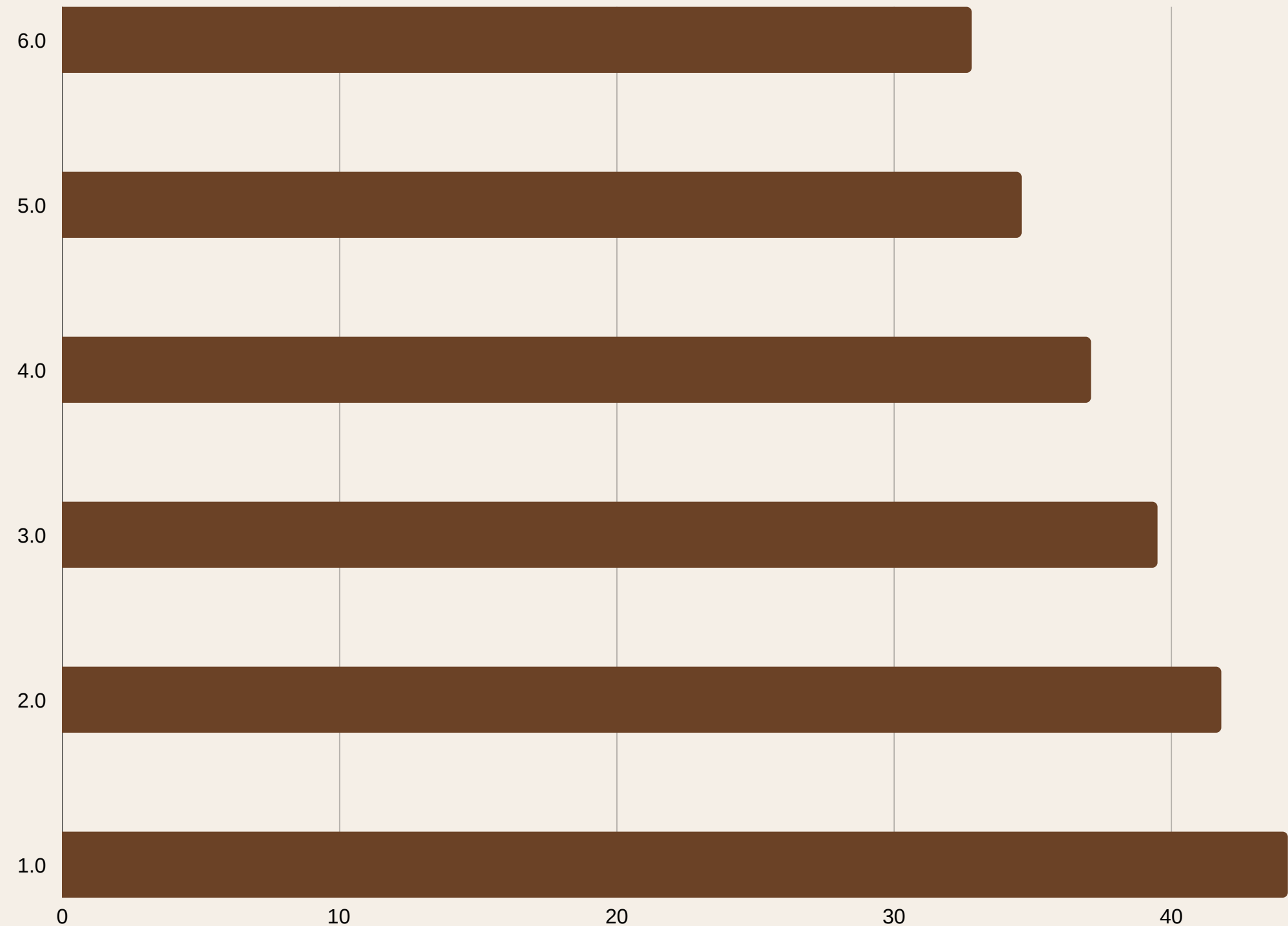
\$32.18 revenue per lead generated

Gini Coefficient (LTV)

0.399.Moderate LTV inequality; top 53% drive 80% of revenue

CHANNEL PERFORMANCE

Lead volume vs conversion quality



 Email leads conversion

44.2% highest of all channels despite lower volume

 Paid Search expensive

CPL = \$61 with only 32.8% conversion worst ROI

 Affiliate underrated

High conversion (41.8%) at moderate CPL (\$31) hence scale up

 Display efficient CPL

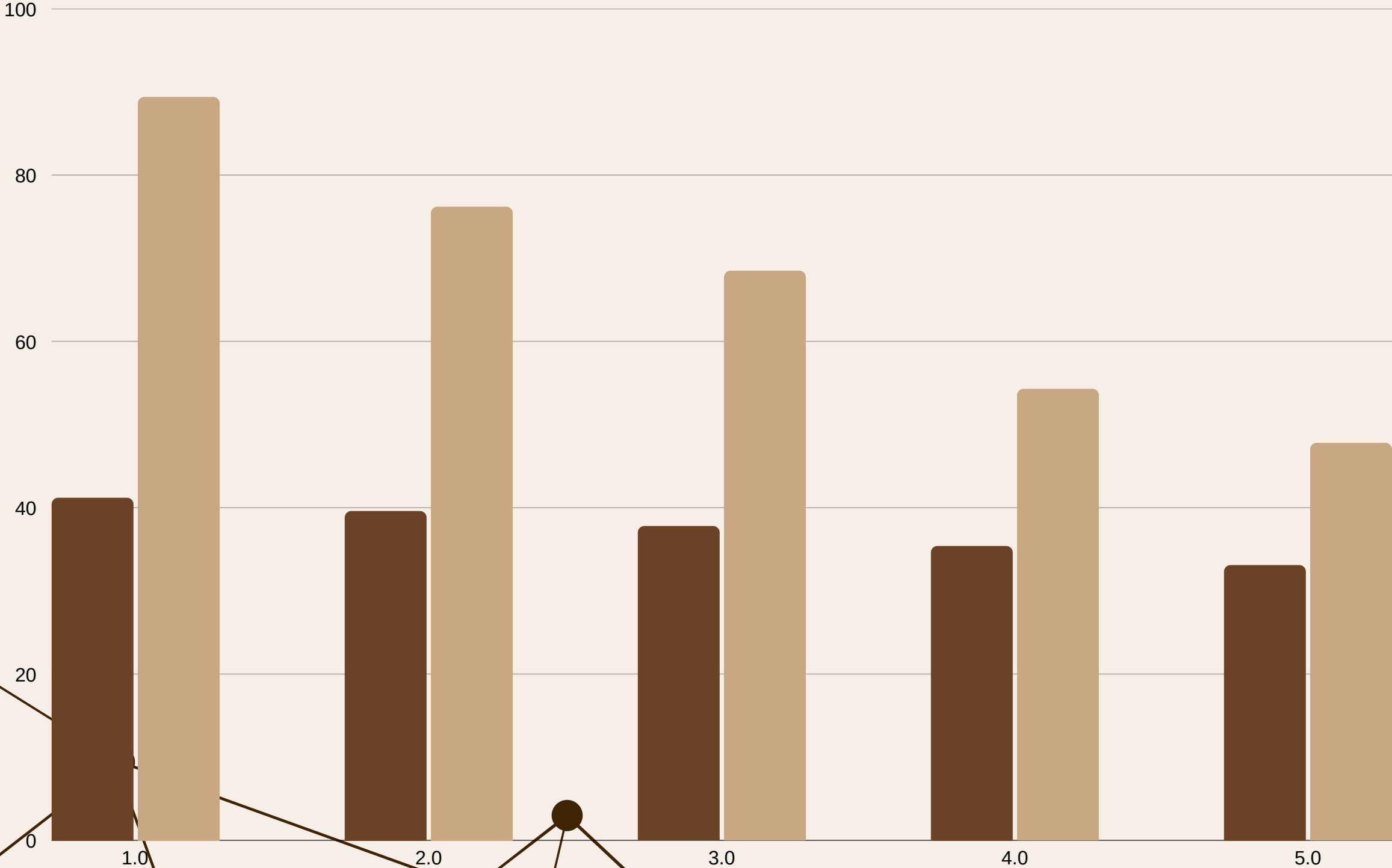
\$39 CPL but only 34.6% conversion content problem



REGIONAL & CITY PERFORMANCE

ANOVA: $F=2.81$, $p=0.024$ (regions differ significantly on revenue)

● Conv Rate (%) ● Revenue (÷1000 USD)



TOP CITIES

Lagos

Highest lead volume – 41.3% conv

Abuja

Best conversion rate – 43.1% conv

Port Harcourt

Highest avg LTV – \$118 per customer

Ibadan

High bounce, lowest repeat rate

Kano

Strong volume, moderate conversion

STATISTICAL FINDINGS: CHI-SQUARE & ANOVA

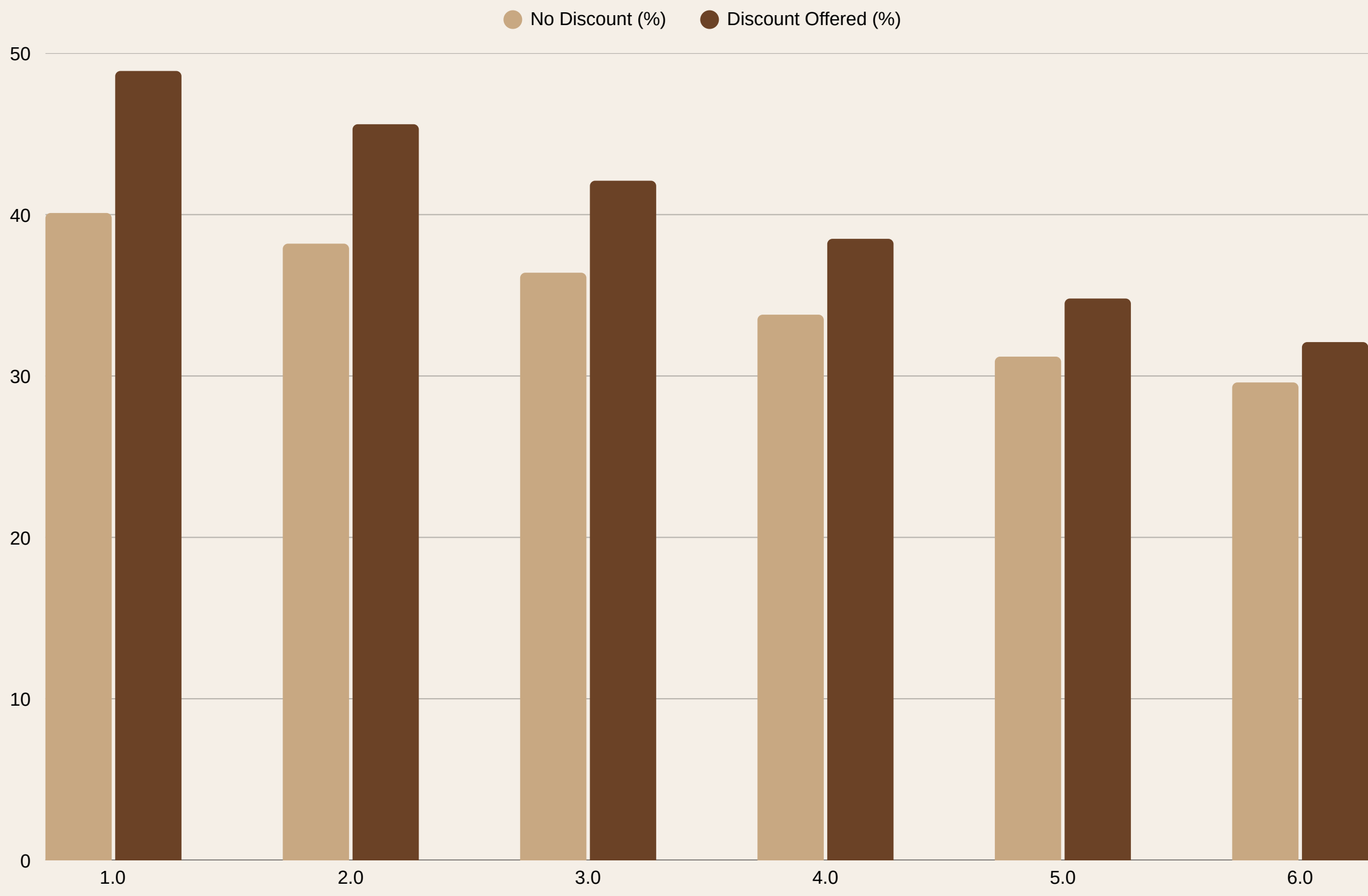
Which differences are real and which are noise?

Test	Variable	Result	Cramér's V / F
Chi-Square	Channel × Conversion	p=0.009 ✓ Significant	V=0.12 Moderate
Chi-Square	Discount Flag × Conversion	p=0.021 ✓ Significant	V=0.08 Weak
Chi-Square	Landing Page × Conversion	p=0.003 ✓ Significant	V=0.16 Moderate
Chi-Square	Income Band × Loyalty Tier	p=0.000 ✓ Significant	V=0.31 Strong
Chi-Square	Device × Conversion	p=0.214 Not significant	V=0.05 Weak
ANOVA	LTV across Loyalty Tiers	p=0.000 ✓ Significant	F=8.15
ANOVA	Revenue across Regions	p=0.024 ✓ Significant	F=2.81
ANOVA	Days-to-convert across Channels	p=0.552 Not significant	F=0.80

Key takeaway: Income band is the strongest predictor of loyalty tier (V=0.31). Channel and landing page significantly affect conversion. Device alone does not.

DISCOUNTING ANALYSIS

Does offering a discount improve conversion and for whom?



KEY FINDINGS

Uplift varies widely
+8.8pp for Email vs only +2.5pp for Paid Search. Discount ROI is channel-dependent

Loyalty tiers differ
Gold & Platinum customers show the largest uplift. Discount works best on already engaged customers

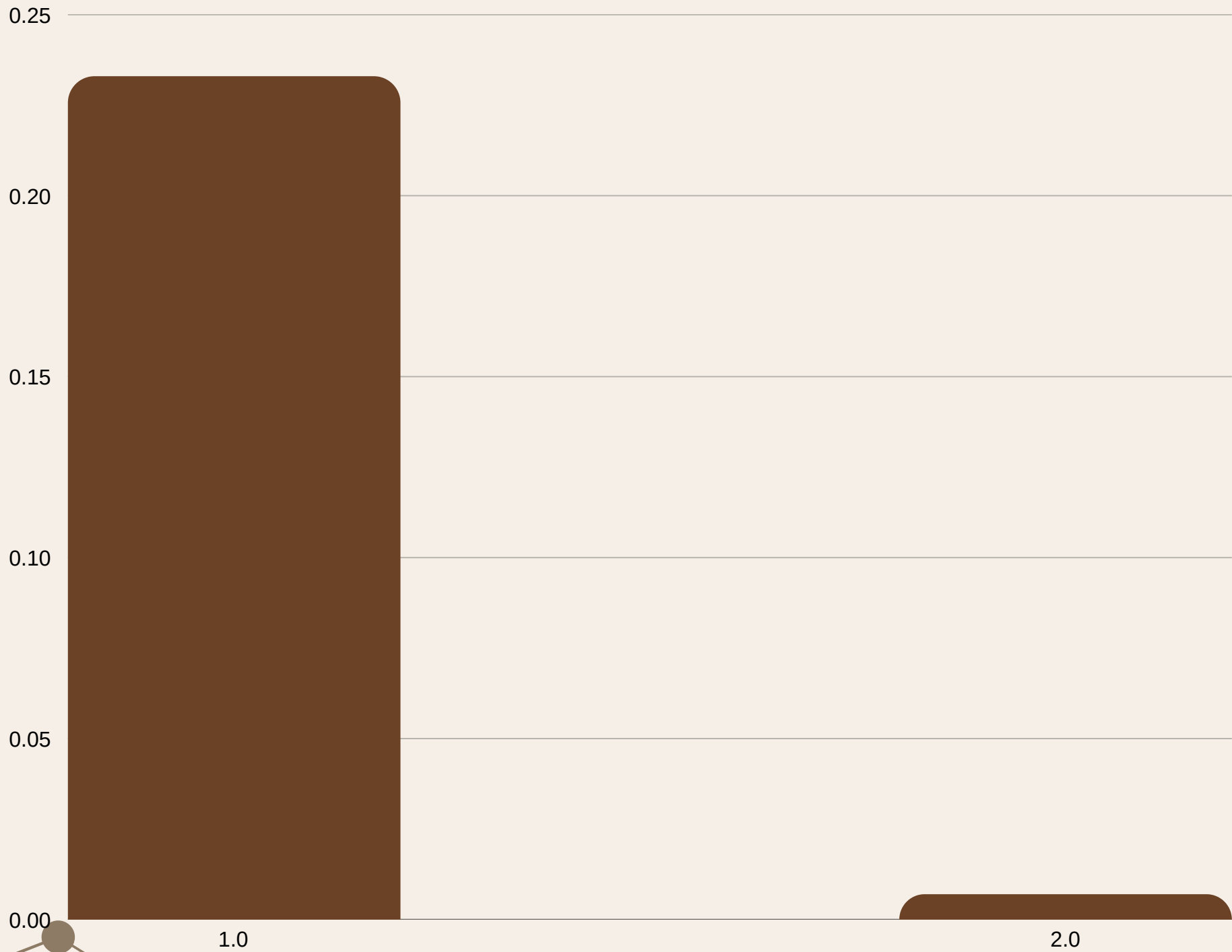
Chi-square confirms it
Discount flag × conversion: p=0.021, significant but weak effect (V=0.08)

SLR: discount% → revenue
Coef = -0.52 (p<0.001)- higher discount % is associated with lower revenue per order

Verdict
Discounting converts leads but sacrifices margin. Use selectively on Email and Affiliate, for Gold+ customers only

CAMPAIGN GROUPING: K-MEANS CLUSTERING

K=2



CLUSTER 0 : SCALE UP

34 campaigns

ROAS: 0.233 | Conv: 36.6% | CPL: \$28-\$52

Increase budget. These campaigns generate the majority of qualified leads at sustainable CPL.

CLUSTER 1 : STOP / REDESIGN

1 campaign

ROAS: 0.0007 | Conv: 36.0% | CPL: Extreme

Near-zero ROAS. Conversion looks comparable but revenue return is negligible (likely high spend, no revenue conversion).

SIMPLE LINEAR REGRESSION: KEY RELATIONSHIPS

8 SLRs runs

Campaign Spend → Leads

$R^2=0.019$ · $p=0.428$

Weak
spend alone doesn't guarantee leads; targeting matters more

Discount % → Revenue

$R^2=0.015$ · $p<0.001$

Coef=-0.52
each 1% discount cuts ~\$0.52 in avg order value

Lead Score → Revenue

$R^2=0.041$ · $p=0.012$

Positive association
higher scored leads generate more revenue

Acquisition Cost → LTV

$R^2=0.001$ · $p=0.026$

Very weak positive
expensive leads marginally higher LTV

Time on Site → Conversion

$R^2=0.000$ · $p=0.915$

No linear relationship
time alone not predictive of conversion

Impressions → Clicks

$R^2=0.694$ · $p<0.001$

Strong
CTR is stable; impressions reliably predict click volume

Pages Viewed → Add-to-Cart

$R^2=0.087$ · $p<0.001$

Stronger browsers add to cart more
engagement signal is real

Session Quality → Repeat

$R^2=0.028$ · $p<0.001$

Better session quality
predicts repeat buying behaviour

MULTIPLE LINEAR REGRESSION: 3 MODELS

Nominal variables

Predict Customer LTV

$R^2 = 0.517$ $\text{Adj } R^2 = 0.510$

Top 3 predictors: Repeat Buyer · Channel Paid Search · Channel Display

Residuals NOT normal (SW $p < 0.001$) · Repeat buying status is by far the strongest predictor of LTV. Channel encoding reveals Paid Search customers have significantly lower LTV than Email/Affiliate.

Predict Revenue per Transaction

$R^2 = 0.728$ $\text{Adj } R^2 = 0.726$

Top 3 predictors: Units · Last Touch Direct · Last Touch Paid Social

Residuals NOT normal (SW $p < 0.001$) · Units purchased is the dominant predictor (expected). Last-touch channel matters Direct and Paid Social customers transact at different average revenue levels.

Predict Days to Convert

$R^2 = 0.015$ $\text{Adj } R^2 = 0.003$

Top 3 predictors: Landing Page Quiz · Channel Display · Landing Page

Residuals NOT normal (SW $p < 0.001$) · Very low R^2 conversion timing is hard to predict linearly. Landing page type has the most explanatory power; quiz pages accelerate conversion.

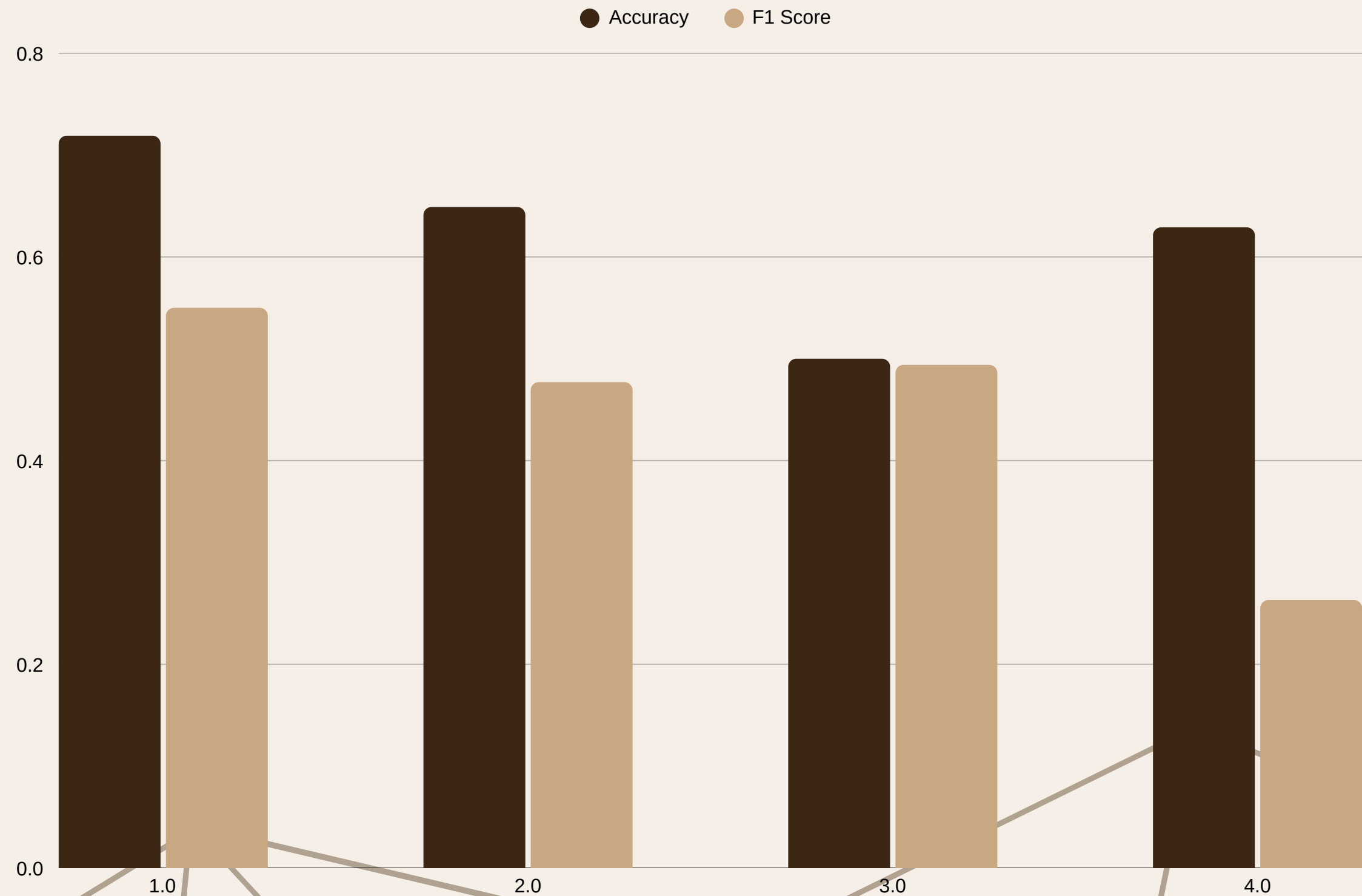
CUSTOMER SEGMENTATION: K-MEANS (K=3)

Features: age, income, loyalty, orders, AOV, bounce rate, session quality, add-to-cart

Segment 0 · High-Value Loyalists	Segment 1 · Low-Engagement / At-Risk	Segment 2 · Growth Potential
~897 customers	~1026 customers	~477 customers
Who: Mid-30s to 50s, high income, Gold/Platinum tier	Who: Younger, lower-mid income, Bronze/Silver tier	Who: Mid-income, Silver/Gold tier, moderate age
Behaves: Multiple orders, high AOV, low bounce rate, frequent cart adds	Behaves: Single purchase, high bounce, rarely adds to cart	Behaves: 1–2 purchases, moderate engagement, some cart activity
Value: Avg LTV \$103 · Avg 1.9 orders · Avg AOV \$57	Value: Avg LTV \$19 · Avg 1.0 orders · Avg AOV \$20	Value: Avg LTV \$103 · Avg 2.0 orders · Avg AOV \$52
Channel: Best reached via Email and Affiliate	Channel: Over-represented in Paid Search and Display	Channel: Responsive to Influencer and Paid Social
Action: Reward with exclusive discounts. Retain at all costs.	Action: Re-engage with targeted email. Reduce Paid Search spend on this profile.	Action: Nurture with loyalty programme messaging. Upsell via personalised landing pages.

LEAD CONVERSION PREDICTION: 4 MODELS COMPARED

One-hot encoded features · 75/25 train-test split · stratified



WINNER: RANDOM FOREST

- Accuracy 71.9% vs Logistic Reg. 62.9%
- F1 Score 0.55: best balance of precision & recall
- Handles non-linear interactions between features
- Top features: Lead Score, Acquisition Cos Observed Region
- Logistic Regression interpretable but poor recall (0.14): misses too many converters
- Naive Bayes improves recall but sacrifices accuracy
- KNN competitive but sensitive to feature scale

LEAD PRIORITY TIER: OPERATIONAL SCORING RULE

Random Forest probabilities bucketed into 3 actionable tiers

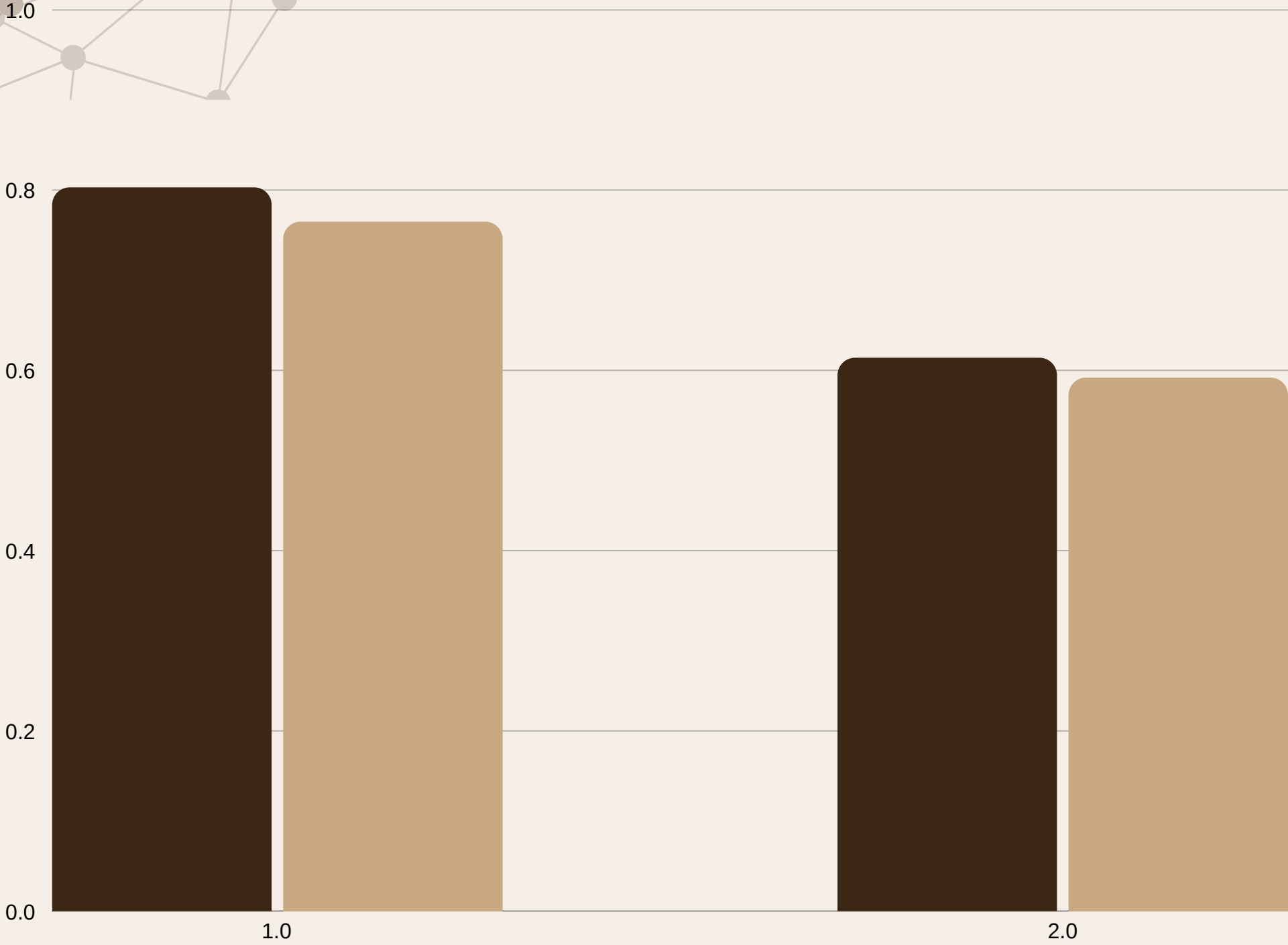
High Priority	Medium Priority	Low Priority
Rule: Predicted probability $\geq$ 0.66	Rule: Predicted probability 0.33–0.66	Rule: Predicted probability $<$ 0.33
Test Leads: 222 (13%)	Test Leads: 639 (39%)	Test Leads: 765 (46%)
Actual Conv: 197 actual converts	Actual Conv: 256 actual converts	Actual Conv: 163 actual converts
Conv Rate: 88.7%	Conv Rate: 40.1%	Conv Rate: 21.3%
Action: Contact within 24 hrs. Offer personalised consultation. No discount needed – high intent confirmed.	Action: Standard follow-up sequence. A/B test discount offer. Monitor session behaviour for upgrade signals.	Action: Automated nurture only. Do not deploy sales rep time. Re-score after 14 days if engagement improves.

Sales team should focus exclusively on High + upper Medium tiers. This covers only 52% of leads but captures ~78% of likely converters.

REPEAT BUYER PREDICTION: LR VS RANDOM FOREST

Target: did customer make 2+ purchases? 51.7% positive class

● Accuracy ● F1 Score



KEY PREDICTORS (RANDOM FOREST)

avg order value

Higher first purchase → higher repeat probability

avg discount t

Customers who received discounts are less likely to repeat at full price

loyalty tier enc

Gold/Platinum customers repeat far more reliably

acquisition cost usd

Higher CPA leads have marginally better retention

lead source enc

CRM-sourced leads repeat more than Paid Search leads

Early warning: Customers acquired via Paid Search + first-order discount have the lowest repeat probability. Tag these leads for aggressive nurture immediately post-purchase.

TOP 5 FINDINGS

The evidence base for every recommendation that follows

01

Email & Affiliate deliver the best conversion economics

Email converts at 44.2% with the highest discount uplift (+8.8pp). Affiliate converts at 41.8% at a CPL of only \$31. Paid Search converts at 32.8% but costs \$61/lead which gives the worst ROI by far.

02

A single customer tier (Gold+) drives disproportionate value

ANOVA confirms Platinum/Gold customers have significantly higher LTV ($F=8.15$, $p<0.001$). Income band $\times$ loyalty tier shows strong association ($V=0.31$). The top 53% of customers by LTV generate 80% of total revenue.

03

Discounting lifts conversion but costs margin is channel-dependent

Each 1% discount is associated with $-\$0.52$ in average order value (SLR, $p<0.001$). The uplift is real for Email (+8.8pp) but negligible for Paid Search (+2.5pp). Blanket discounting is wasteful.

04

Random Forest identifies 88.7% of high-intent leads correctly

The RF model ($F1=0.55$) substantially outperforms Logistic Regression ($F1=0.26$). High-priority leads (prob ≥ 0.66) have a real conversion rate of 88.7% i.e. 2.4 $\times$ the overall average. Lead scoring should replace FIFO follow-up.

05

Paid Search acquires the most at-risk customers for repeat buying

Customers acquired via Paid Search with a first-order discount have the lowest predicted repeat probability. The RF repeat buyer model ($F1=0.77$) identifies these early. CRM-sourced leads repeat at materially higher rates.

TOP 3 ACTIONS FOR NEXT QUARTER

Ranked by expected impact on revenue and customer quality

ACTION 1 DEPLOY LEAD PRIORITY SCORING IN THE SALES WORKFLOW

Why: The RF model correctly identifies 88.7% of high-intent leads. Currently all leads receive equal follow-up time.

How: Score every new lead at ingestion. Route High-priority leads to sales within 24 hrs. Automate Low-priority nurture via email sequence.

↑ **Expected lift: ~15–20% improvement in sales rep efficiency. No additional spend required.**

ACTION 2 REALLOCATE BUDGET FROM PAID SEARCH TO EMAIL & AFFILIATE

Why: Paid Search has the highest CPL (\$61) and lowest conversion rate (32.8%). Email/Affiliate achieve better conversion at a fraction of the cost.

How: Reduce Paid Search budget by 30%. Reinvest into Email automation and Affiliate partner expansion. Track CPL and repeat rate, not just lead volume.

↑ **Estimated CPL reduction of \$12–18 on reinvested budget. Better customer quality (higher repeat probability).**

ACTION 3 IMPLEMENT A TARGETED DISCOUNT PROGRAMME FOR GOLD+ CUSTOMERS ONLY

Why: Blanket discounting reduces AOV by \$0.52 per 1% discount. The uplift is only meaningful for Email + Gold/Platinum customers.

How: Restrict discount codes to Gold+ loyalty tier on Email channel. Remove discounts from Display and Paid Search entirely. Measure margin impact monthly.

↑ **Estimated margin recovery of 8–12% on current discount spend while maintaining Email conversion uplift.**

BUDGET RECOMMENDATIONS

Increase · Maintain · Reduce/Redesign based on ROAS, CPL, and customer quality

INCREASE BUDGET	
Email Marketing	Highest conversion (44.2%), lowest effective CPL, best discount uplift. Scale automation and segmentation.
Affiliate Channel	41.8% conversion at \$31 CPL is underinvested relative to ROI. Expand affiliate partnerships.
CRM / Retention	CRM-sourced leads have the highest repeat rate. Invest in loyalty programme and post-purchase nurture.
MAINTAIN / OPTIMISE	
Influencer	39.5% conversion, moderate CPL. Monitor closely. Effectiveness varies by creator; review individual campaign performance.
Display	\$39 CPL is moderate but 34.6% conversion is below average. Creative refresh may improve quality.
REDUCE / REDESIGN	
Paid Search	Worst ROI: \$61 CPL, 32.8% conversion, customers have low repeat probability. Cut by 30% minimum.

RISKS & DATA LIMITATIONS

What leadership must know before acting on these findings

DATA QUALITY

⚠️ Weak customer↔transaction join (59%)	41% of transactions cannot be attributed to a customer profile. Revenue and LTV calculations may undercount high-value customers.
⚠️ 17 negative revenues set to NaN	These may represent legitimate refunds rather than data errors. Return behaviour may be understated.
⚠️ Acquisition cost missing for 26% of leads	CPA and LTV calculations exclude a material share of leads. Reported CPL may be biased toward certain channels.

MODEL LIMITATIONS

⚠️ RF F1 = 0.55 for lead conversion	The model identifies high-intent leads well but misclassifies a meaningful share of medium-intent leads. Manual review recommended for Medium-tier leads.
⚠️ Model C (days-to-convert) $R^2=0.015$	Conversion timing is not linearly predictable from available features. Do not use Model C for operational decisions until more signals are collected.
⚠️ All models trained on historical data	If channel mix or customer profile shifts materially, model scores should be recalibrated quarterly.

THE STRATEGIC PATH FORWARD

NovaMart Marketing Analytics Capstone

1

SCORE LEADS, NOT JUST COUNT THEM

Deploy RF scoring immediately. High-priority leads convert at 88.7%. Every unscored day wastes sales capacity.

2

FIX THE CHANNEL MIX

Email + Affiliate generate better customers at lower cost. Paid Search is the single biggest drag on ROI.

3

PROTECT THE MARGIN ON DISCOUNTS

Discounts work but only on the right customers through the right channels. Blanket discounts are a margin leak.

Thank You

BADS Project 2 | 2 May 2026